Department of Computer Science and Engineering

Academic Year 2021 - 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE 'A'
Course Code & Title: EC8393 & Fundamentals of Data Structures in C
Name of the Faculty member (s): Ms.P.Jothi Thilaga, AP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit V / Sorting Techniques
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO13
- **Activity Chosen:** Flash Cards
- **Justification:**

  In unit V, the students need to understand various sorting techniques such as insertion sort, bubble sort, merge sort and quick sort. Also they need to understand the technique used and complexity of each sorting technique. This activity helps the students to understand the techniques and its complexity clearly.

- **Time Allotted for the Activity:** 30 Minutes

- **Details of the Implementation:**

  A flashcard is a method in which the questions and corresponding answers are in separate cards. Question and answer cards are prepared on sorting techniques as shown in Figure 1, taught in the previous day's class. Initially, Flashcards containing questions and answers were distributed to students in the class. One student from the class displayed the question card, the other students in the class identified the cards relevant to the questions within a minute. Then the students explained the answer in front of the classroom for the next 2 minutes with their answer cards, relevant to the corresponding question as shown in Figure 2. This activity helped the students to revise the different sorting techniques in unit V.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PSO1
CO5	3	3	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PSO1
	3	3	3	2	2
Justification for correlation	Needs engineering and mathematical knowledge for understanding sorting concept	Able to identify relevant sorting, searching problem for the given applications	Able to interpret the complex engineering problems with the help of the sorting algorithms	Research-based knowledge will help to identify the suitable sorting and searching algorithms.	Able to implement different sorting and searching algorithms in IT applications

- **Images / Screenshot of the practice:**

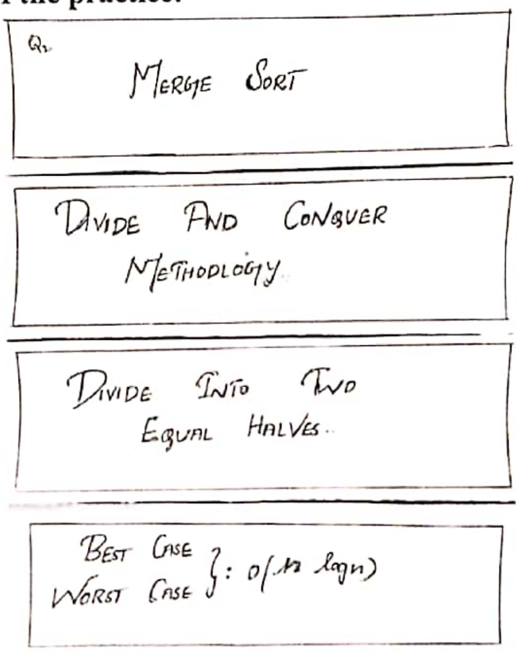

Figure 1: Flash Cards

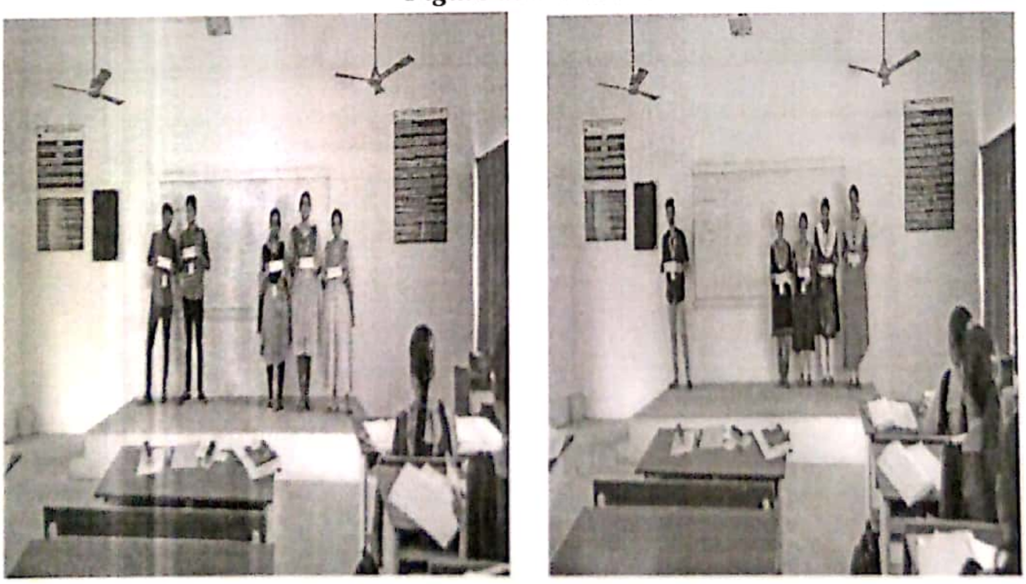

Figure 2: Flash Card Implementation

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - All the students were actively participated and enjoyed once they identify answered for the concepts correctly.
    - Students explored the knowledge of the various sorting techniques.
    - It makes the students to gain well knowledge on different sorting and searching techniques concepts, complexity and applications.
  - ***Benefit of the practice:***
    - Students can able to write the sorting techniques questions in Internal Test.
    - Students can able to perform different sorting techniques for any applications.
    - They can able to identify the corresponding sorting technique for the applications and implement using C programming.
    - This activity helps the students to attend the sorting technique questions in the examination
  - ***Challenges faced in implementation:***
    - Few students can get difficulties in identifying the complexity of the different sorting algorithms.
    - Some students were hesitated to express the relevant answers to the question card.

References:

- https://www.teachingenglish.org.uk/article/using-flash-cards-young-learners
- https://eltexperiences.com/flashcards-in-the-classroom-ten-lesson-ideas/
- https://www.ritrjpm.ac.in/images/computer-science/FlashCards-Venkat-INP.pdf

24/11/2021
Signature of Faculty Member

24/11/2021
HOD